A. Venkata Vinesh Kumar Reddy

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Education

Mahindra University, School of Engineering — Hyderabad, India
Bachelor of Technology in Computer Science and Engineering; CGPA: 7.96/10 — Aug 2023 – Present

HMR International — Bengaluru, India
Intermediate Education (PCM); Percentage: 80% — June 2020 – May 2022

Narayana Olympiad School (CBSE) — Bengaluru, India
10th Grade Secondary School Certificate; Percentage: 91.2% — May 2019 – June 2020

Technical Skills

Programming Languages: Python (Primary), C++, SQL, MATLAB
Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, NumPy, Pandas, LSTM, GRU
Software Engineering & Tools: FastAPI, React, Docker, Git, REST APIs, JSON
Mathematical Foundations: Probability, Statistics, Optimization, Time Series Analysis, Reinforcement Learning (Q-Learning)

Technical Projects

Asset Price Prediction Platform — Jan 2026 – Present
Developer; Stack: FastAPI, React, PyTorch, Statsmodels — github.com/VenkataVinesh/Asset-Price-Prediction-Platform

- Built an interactive forecasting platform to evaluate how sequential deep learning models (LSTMs) compare with classical statistical baselines (ARIMA) on noisy pricing data.
- Created data pipelines to calculate 14-day rolling technical features (SMA, EMA, RSI, and Volatility indicators) in Pandas.
- Developed a FastAPI backend to run model inference and serve real-time predictions to a responsive React dashboard.
- Integrated interactive chart visualizations displaying historical asset prices against projected multi-step forecast intervals.

Weather Forecasting Analytics — June 2025 – July 2025
ML Developer; Stack: Python, Pandas, Statsmodels, PyTorch — github.com/VenkataVinesh/Weather-Forecasting-Analytics

- Analyzed multi-dimensional meteorological sensors, implementing ARIMA, SARIMA, and PyTorch LSTM networks to forecast temperature and humidity.
- Conducted time-series decomposition to isolate trend and seasonal patterns, adjusting model hyperparameters based on residual stationarity checks.
- Compared sequence network architectures against statistical baselines, documenting computational latency and forecasting accuracy trade-offs.

Veltrix — Algorithmic Trading Dashboard — Dec 2025 – Present
Developer; Stack: Next.js, TypeScript, Python, FastAPI, Docker — github.com/VenkataVinesh/Veltrix

- Aggregated historical daily stock price series and computed asset allocations using Markowitz/Sharpe optimization solvers without introducing interface rendering lag.
- Programmed a Next.js dashboard using Recharts to plot price histories and expected portfolio variance. Integrated a FastAPI (Python) backend using SciPy solvers to compute optimal portfolio weights.
- Containerized and deployed the full-stack system using Docker, providing responsive visualization of strategy backtests and mathematical frontiers.

Certifications

Deep Learning Specialization (Coursera) – Comprehensive training in Neural Networks, Backpropagation, CNNs, and RNNs/LSTMs using TensorFlow.

Leadership & Extra-Curriculars

Math Club, Mahindra University — Hyderabad, India
Logistics Head — Aug 2024 – Present

- Organized logistics and scheduling for university math modeling competitions, academic events, and problem-solving seminars.
- Coordinated resource management, venue setup, and schedules across departments to support academic activities.

TEDx Mahindra University — Hyderabad, India
On-Ground Operations Lead — Jan 2024 – May 2024

- Coordinated on-ground event operations, venue setup, and logistical coordination for speakers and over 500 attendees.
- Managed event flows, speaker timelines, and coordinated team communication for event execution.

Smart India Hackathon (SIH) — National Level Competition
Participant & Team Developer — Sept 2024 – Oct 2024

- Collaborated in a team of 6 to prototype an intelligent system solution, assisting in solution ideation and backend API prototyping.